



## How far footballers run?

Watching a football game is fun, but have you wondered how much do the players run in total? Find out here <http://dgit.in/1p9p9HK>



## Easier rooting for Android

Rooting your Android phone is tedious and with Towelroot Exploit you could do it right from your web browser <http://dgit.in/1nXLPXZ>

Workshop

Simply start out by cutting all wires down to the same length. There have to be 26 pieces of wires which are about 10 centimetres long. You can make them longer or shorter if you wish to route the cables according to your needs. Expose about 5mm of the wire on each end. Use the ATX connector toolkit to crimp the metal pins onto each wire. Ensure that the male and female ports are aligned the right way. You need to put an extra wire into pins 16 and 17 each for one female port. To make things clear.

- One female port to be connected to the male port pin-to-pin (Port 1 to Port 2)
- Pin 16 of the same female port is to be connected to pin 16 of the other female port (Port 1 to Port 3)
- Pin 17 of the same female port is to be connected to pin 17 of the other female port (Port 1 to Port 3)

When you're done the arrangement must look like the image below.



The finished project should look like this

All that's left to do now is to insert the 24-pin connectors from both power supplies into the contraption you just made. Then attach the male connector to your motherboard. Ideally, you should connect half the peripherals to one power supply and the rest to the other. Often, power supplies are designed such that load regulation mechanisms will cause the voltage level on one rail to be lower than what it should be if all rails are not loaded. Check the product documentation to figure out if this is the case for your SMPS. If yes, then you'll need to attach the two 10 Ohm resistors in parallel first, then connect the leads between 5 volts (red) and ground (black) on any cable of the power supply that isn't connected to the motherboard. We put two resistors in parallel so that the current is divided between the two resistors and the



Add2PSU comes with a relay switch

heat generated is also dissipated between the two resistors.

## 2. Third party solutions

Add2Psu is a simple connector that does the job of connecting two power supplies for you. All you need to do is insert the 24-pin connector of one power supply onto the board and a 4-pin molex connector of the other power supply onto the Add2Psu board.

The board is quite simple in design. The 4-pin molex connector triggers a relay switch which in turn short-circuits pin 16 and pin 17 of the 24-pin ATX connector, thus, switching it on as soon as the first power supply is turned on. Using this board you can chain as many PSUs as

you want. We've seen this being used extensively by folks mining cryptocurrency but three power supplies are the maximum that we've seen chained together in the system.

## 3. Specially designed PSUs

The last option is to buy power supplies that have been designed to be linked to each other. Antec's HCP (High Current Pro) series come with a feature called OC Link. Using this feature one can connect two such power supplies together without worrying about any parameters like load balancing rails or worrying about frequency tolerance.

Do take care while making this project and test the adaptor that you've made for continuity using a multimeter before you connect the device to your two power supplies. Ideally, it is preferred that you load the entire computer on one power supply and all the graphics cards on the other. For cryptocurrency mining ensure that all mining gear are equally distributed across all powersupplies.

We're eager to hear how this workshop turned out for you and if you've found out any other ways of combining two power supplies then do write in and let us know. 



Third-party solutions like Antec's OC link takes all the fuss and fun out of the job